Daniel Boyland

Biomedical Scientist · Clinical Education & Quality Specialist

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Core specialisms

ISO, quality & technical

- ISO 15189, 17025 & 17043 (train, assess, implement)
- QMS governance, documentation & audit readiness
- Technical assessment of laboratories
- Evidence validation & data integrity

Learning, curriculum & pedagogy

- Instructional design & blended learning
- Apprenticeship & RQF curriculum (L3–L6)
- Video scripting, storyboarding & media selection
- Competence frameworks & assessment strategy
- Metacognitive & learning-science models

Education

MSc Biomedical Sciences (Immunohaematology) — Merit
Cardiff Metropolitan University · 2016

BSc (Hons) Biomedical Sciences — First Class
Cardiff Metropolitan University · 2007

BSc Biology — 2:1
University of Portsmouth · 2003

Registration

- HCPC Registered Biomedical Scientist
- Chartered Scientist (CSci)
- IBMS Full Member

Professional profile

Biomedical scientist and clinical education specialist with a strong diagnostic background across haematology, transfusion, coagulation and haemoglobinopathy screening. Over a decade in ISO 15189 and CPA-accredited NHS laboratories, progressing into technical assessment, quality systems and specialist training. Now founder of Forged Frameworks, translating scientific and regulatory rigour into accessible, high-quality learning.

70+	650+	2,000+	20
ISO 15189 & 17025 lab assessments	Training days moved online, Feb 2020 to Mar 2021	Hours logged in the first year of Forged Frameworks, 40+ modules	Line reports at most (NHS); 47 trainers led at UKAS

Professional experience

Biomedical, Diagnostic & Managerial Roles
Feb 2004 – Jul 2018

NHS Wales — Cardiff & Vale UHB

- Trained staff to the IBMS standards of competence and HCPC standards of proficiency, so trainees could gain the IBMS Certificate of Competence and complete Specialist Portfolios in haematology.
- Led the CPA→ISO 15189 transition as a project: change control, method validation, measurement uncertainty, budgets and quality documentation.
- Aligned laboratory training to ISO 15189, BSQR, HTA, JACIE, HCPC and IBMS standards for every grade of staff.
- Built training and competency matrices; planned and assessed placements for trainees on IBMS-accredited degrees; IBMS verifier and examiner.

Training & Standardisation Co-ordinator — Dec 2016 – Jul 2018
Training remit for ~150 staff across 8 sections, from support workers to senior scientists; chaired a 9-lead training team; succession and contingency planning.

Acting Automated Haematology Senior — Apr – Dec 2016
Line-managed about 20 staff (Biomedical Scientists, MLAs, support and clinical staff): 1:1s, appraisals, absence and capability; led the CPA→ISO 15189 transition.

Acting Haemoglobinopathy Diagnostic Service Manager — May 2015 – Jun 2016
Managed a team of 5 (Biomedical Scientists and support staff); Wales-wide screening for 4 of 7 health boards; ~£42k/yr saving via new algorithm.

Specialist Biomedical Scientist — 2009 – 2015

Biomedical Scientist & Medical Laboratory Assistant — 2004 – 2009

Further specialisms

Leadership & collaboration

- Project leadership & cross-functional coordination
- Stakeholder engagement — employers, SMEs, regulators
- Design-workshop facilitation with SMEs & curriculum teams
- Coaching, mentoring & capability development
- Commercial training delivery & consultancy

Line management & people development

- Line management of teams of 2 to 20
- 1:1s, appraisals, objectives & development plans
- Recruitment, onboarding & induction
- Absence, performance & capability
- Succession & contingency planning

Certifications

- TAP Diploma in Learning Design (2021)
- IBMS Certificate of Expert Practice in Training (2017)
- IBMS Specialist Diploma in Haematology & Transfusion (2007)
- UKAS Technical & Lead Assessor — ISO 15189, 22870, 17043

Platforms & tools

Laboratory & diagnostics

Automated haematology · coagulation · transfusion · haemoglobinopathy screening · EQA / IQA

Quality systems

Q Pulse · iPassport · Telepath / Dedalus · Ideagen

Authoring & media

Articulate 360 (Rise, Storyline) · SCORM · xAPI · Visme · Canva · Vyond · Descript · Camtasia

Code & AI

Python · HTML/CSS/JS · Claude Code · OpenAI Codex · OpenAI & Gemini APIs · Google Apps Script · NotebookLM

Professional experience (continued)

Technical Assessor & Quality Specialist

Nov 2017 – 2024

UKAS

Continuous parallel thread alongside training roles — UK & international.

- Conducted 70+ technical assessments across healthcare, proficiency testing, forensics and calibration.
- Advised on competence frameworks, audit preparation and process improvement; led data-integrity checks.
- Supported international accreditation bodies (Saudi Arabia, Ukraine, Netherlands).

Assessment Manager, Medical Laboratories

July 2018 – Feb 2020

UKAS (Healthcare)

- Planned and led ISO 15189 assessments for clinical and diagnostic laboratories, including pan-pathology teams.
- Oversaw assessor competency, evidence review and governance.
- Supported organisations preparing for ISO 15189 and 17025 accreditation.

Senior Training Manager

Feb 2022 – Sept 2024

UKAS Training Academy

- Ran the ISO 15189, 17025 and 17043 programme for clinical, testing and calibration laboratories; managed budgets and external resource and grew the digital and blended portfolio the COVID year had started.
- Line-managed up to 3 training managers and coordinators: 1:1s, appraisals, recruitment, onboarding, absence and development plans.
- Led a pool of about 30 internal and external training facilitators, 47 at its peak, plus SMEs and designers: onboarding, competency and quality of delivery.

Founder & Creator

Aug 2025 – Present

Forged Frameworks

AI-driven, human-assured learning-design consultancy.

- Design and build Level 3 to Level 6 apprenticeship and degree modules and Catalyst framework packages for Tiro Training, mapped to KSBs, RQF and FHEQ outcomes: 1,500+ hours across 28 since September 2025.
- Design HCP interactive education for a medical-education publisher: 150 hours across five oncology, haematology and diabetes-technology programmes since June 2026, from prescribing guidance to decision tools and patient-management pathways.
- Provide WCAG / EN 301 549 accessibility consultancy, ISO quality-system support (Digistain) and AI-literacy / prompt-engineering capability building.

Laboratory specialisms

14 yrs — **NHS pathology**
Medical Laboratory Assistant to acting manager, Cardiff & Vale UHB

Haem — **Haematology & coagulation**
Automated haematology, morphology and coagulation validation; CPA→ISO 15189

Tx — **Blood transfusion**
Transfusion validation and EQA participation

Hb — **Haemoglobinopathy screening**
Wales-wide service for 4 of 7 health boards; ~£42k/yr saving

IBMS — **Training & standards**
IBMS Certificate of Competence and Specialist Portfolios; HCPC, BSQR, HTA, JACIE

20 · 5 — **NHS teams**
Line-managed about 20 in haematology and 5 in haemoglobinopathy; training remit for ~150

Interests

Sci-fi, science and literature. Creative metalwork and blacksmithing — building my own forge and metalwork brand. Walking, running and wellbeing. Mental-health advocacy, and two very opinionated cats.



Professional experience (continued)

Learning Content Manager
Sept 2024 – Aug 2025

Tiro Training Ltd

- Scripted and designed 240+ hours of L3–L6 blended learning — including video scripts and storyboards — across Bioscience, Asbestos, Materials, Chemical Science, Polymer & Processing and Laboratory Management.
- Aligned curriculum to apprenticeship standards and RQF outcomes; contributed to 100% external certification success.
- Redesigned the Moodle VLE and integrated GenAI tools (NotebookLM, ChatGPT) into the content pipeline.
- Line-managed 2 content staff and led about 12 contract SMEs and designers: 1:1s, objectives and development plans; mentored them in instructional design, accessibility and inclusive practice.

Training Manager
Feb 2020 – Feb 2022

UKAS Training Academy

- Moved 650+ training days of classroom teaching to remote delivery between Feb 2020 and March 2021: material rebuilt for Adobe Connect, Rise 360 learning packs and pre-packs around each session.
- Developed healthcare content: measurement uncertainty, PoCT, proficiency testing, ISO 19011 audit.
- Supported assessor-workforce competency, documentation and digital delivery.

Founder & Blacksmith
2020 – Present

Wig's Wonky Forge

Continuing professional development

GenAI in Learning Design (Coursera / Google) · Accessibility & Universal Design for Learning · ISO 15189:2022 Implementation & Internal Audit · Prompt Engineering for L&D · AI Literacy for Educators.

Values & soft skills

Patient-focused · Scientifically rigorous · Evidence-led · Supportive trainer · Calm under pressure · Collaborative · Integrity-driven